

DTSA 5516

Fair Machine Learning: Algorithms

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	ML Engineer (92.6 / 100)	82.8 / 100
SPECIALIZATION Fair Machine Learning		

83

CURRENT AVG SCORE
across 6 job roles

98

POTENTIAL SCORE
with recommended additions

+24

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course will build upon the student’s new foundational knowledge of the space by introducing key technical aspects for implementation of Fair ML. Students will learn how to measure fairness quantitatively, calculate common fairness metrics, implement mitigation algorithms, and test for significance after mitigating for fairness. Students will learn these technical concepts in context of the Machine Learning lifecycle so they are prepared to address fairness via common pre-processing, intrinsic modeling, or post-processing techniques.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	92.6	100.0	+7.4	LOW
AI Engineer	91.3	100.0	+8.7	LOW
Data Scientist	86.7	100.0	+13.3	LOW
Machine Learning Engineer	80.1	99.4	+19.3	MEDIUM
MLOps Engineer	76.5	100.0	+23.5	MEDIUM
Applied Scientist	69.8	89.5	+19.7	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Business Communication 1 of 6 roles

insights

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 **HIGH** **Add System Design coverage** — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

- 2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: insights.

- 3 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

ML Engineer

92.6 → 100.0 (+7.4 pts)

data systems models problems impact product global building

AI Engineer

91.3 → 100.0 (+8.7 pts)

following data models systems problems technology detection information

Data Scientist

86.7 → 100.0 (+13.3 pts)

data sets product insights different behaviors problems

Machine Learning Engineer

80.1 → 99.4 (+19.3 pts)

data interfaces sets tools design science models applications

MLOps Engineer

76.5 → 100.0 (+23.5 pts)

detection data text systems image training tools models

Applied Scientist

69.8 → 89.5 (+19.7 pts)

problems models data systems industry design regulatory science

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.